

2. Suppose the process of exchanging points for gifts depends on three subsystems in a shopping website, including payment subsystem, integral calibration subsystem, logistics subsystem. The functions of these subsystems are deduct points, verify whether the points pass and enter the distribution process respectively. The name of the facade class is GiftExchangeService, and the name of client program is Test.

(Hint: Use PointsGift as a gift entity)

GiftExchangeService

```
package experiment9_2;

class GiftExchangeService {

    PaymentSystem paymentSystem;

    IntegralCalibrationSystem integralCalibrationSystem;

    LogisticsSystem logisticsSystem;

    public GiftExchangeService(){

        paymentSystem = new PaymentSystem();

        integralCalibrationSystem = new
IntegralCalibrationSystem();

        logisticsSystem = new LogisticsSystem();

    }

    public void exchangeGift(PointsGift gift, int userPoints) {

        System.out.println("Starting gift exchange process for:
" + gift.getName());

        if(integralCalibrationSystem.verifyPoints(gift)){
```

```

        if(paymentSystem.deductPoints(gift,
userPoints)){

            logisticsSystem.dispatchGift(gift);

        }else{

            System.out.println("Gift exchange failed due
to insufficient points.");

        }

    }else{

        System.out.println("Gift exchange failed due to
invalid integral calibration.");

    }

}

}

```

IntegralCalibrationSystem

```

package experiment9_2;

class IntegralCalibrationSystem {

    public boolean verifyPoints(PointsGift pointsGift){

        System.out.println("Verifying points for gift: " +
pointsGift.getName());

        return pointsGift.getPoints() >= 0;

    }

}

```

```
}  
}
```

LogisticsSystem

```
package experiment9_2;  
  
class LogisticsSystem {  
    public void dispatchGift(PointsGift pointsGift) {  
        System.out.println("Entering logistics process for gift:  
" + pointsGift.getName());  
        System.out.println("Gift dispatched successfully!");  
    }  
}
```

PaymentSystem

```
package experiment9_2;  
  
class PaymentSystem {  
    public boolean deductPoints(PointsGift gift, int userPoints)  
{  
        System.out.println("Deducting " + gift.getPoints() + "  
points for gift: " + gift.getName());  
        System.out.println(gift.getName() + " still has " +  
(userPoints - gift.getPoints()) + " points left.");  
        if (userPoints >= gift.getPoints()) {
```

```

        return true;
    } else {
        System.out.println("Insufficient points! Required:
" + gift.getPoints() + ", Available: " + userPoints);
        return false;
    }
}
}

```

PointsGift

```

package experiment9_2;

class PointsGift {
    private String name;
    private int points;

    public PointsGift(String name, int points) {
        this.name = name;
        this.points = points;
    }

    public String getName(){
        return name;
    }
}

```

```

    }

    public int getPoints(){
        return points;
    }

    public void setPoints(int points){
        this.points = points;
    }
}

```

Test

```

package experiment9_2;

public class Test {
    public static void main(String[] args) {
        PointsGift pointsGift = new PointsGift("Lucy", 200);
        GiftExchangeService giftExchangeService = new
GiftExchangeService();

        int userPoints = 300;

        giftExchangeService.exchangeGift(pointsGift,
userPoints);
    }
}

```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA 20
Starting gift exchange process for: Lucy|
Verifying points for gift: Lucy
Deducting 200 points for gift: Lucy
Lucy still has 100 points left.
Entering logistics process for gift: Lucy
Gift dispatched successfully!

Process finished with exit code 0
```